NITRO COMPOUNDS

Chemical properties of Nitro alkane (RNO_2)

Reduction

$R - NO_2$ → (Sn / HCl, acid) → RNH_2

$R - NO_2$ → (Zn / NH_4Cl, Neutral) → $RNHOH$

$R - NO_2$ → (Catalyst) → $R - NH_2$

Hydrolysis

$RCH_2NO_2 \xrightarrow[\Delta]{HCl / H_2O} RCOOH$

$R_2CHNO_2 \xrightarrow{HCl / H_2O} R_2CO + N_2O + H_2O$ (Ketone)

Halogenation

$CH_3NO_2 \xrightarrow[NaOH]{Cl_2} CCl_3NO_2 + 3\,HCl$

Chloropicrin

Methods of preparation of $R - NO_2$ and RONO

From alkyl halides

$R - X$ → ($2\,AgNO_2$) → RNO_2 (Nitroalkane, Major) + $RONO$ (Alkylnitrite, Minor) + $2AgI$

$R - X$ → (KNO_2) → RNO_2 (Minor) + $R - O - N = O$ (Major)

Nitration of alkane

$CH_3 - CH_3 \xrightarrow{HNO_3} CH_3CH_2NO_2 + CH_3NO_2$

Nitration of arenes

$C_6H_6 + HNO_3 \xrightarrow{Con\ H_2SO_4}$ Nitrobenzene ($C_6H_5NO_2$)

$HNO_3 + H_2SO_4$ nitratingmixture

NO_2^+ electrophile

Electrophilic Substituion Reaction

Nitrogroup is meta - directing

Nitration: $C_6H_5NO_2 \xrightarrow{Conc.HNO_3 + H_2SO_4}$ m - dinitobenzene

Sulphonation: $C_6H_5NO_2 \xrightarrow{Conc.H_2SO_4}$ m - nitrobenzene sulphonic acid (SO_3H)

Halogenation: $C_6H_5NO_2 \xrightarrow{Cl_2 / Fe}$ m - chloronitrobenzene

Nitrobenzene (oil of mirabane) – reduction

Nitrobenzene ($C_6H_5NO_2$)

Acidmedium, Sn / HCl → $C_6H_5NH_2 + H_2O$ (Aniline)

Neutral, $Zn + NH_4Cl$ → C_6H_5NHOH (N - Phenylhydroxylamine)

Alkaline medium, $Na_3AsO_3/NaOH$ → $C_6H_5 - N = N - C_6H_5$ (Azoxybenzene)

Alkaline medium, $Zn / NaOH / CH_3OH$ (or) $LiAlH_4$ → $C_6H_5 - N = N - C_6H_5$ (Azobenzene)

Alkaline medium, $Zn / NaOH$ → $C_6H_5 - NH - NH\,C_6H_5$ (Hydrazobenzene)

Electrolytic reduction, Weaklyacidic, Medium → $C_6H_5NH_2$ (Aniline)

Electrolytic reduction, Stronglyacidic, $Conc\ H_2SO_4$ → $H_2N - C_6H_4 - OH$ (P - nitrophenol)

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